EMERGE AI HACKATHON · CHALLENGE 9

genieLOT

Counting cancer treatments the right way

Patrick Charles · github.com/Pchar046/genieLOT

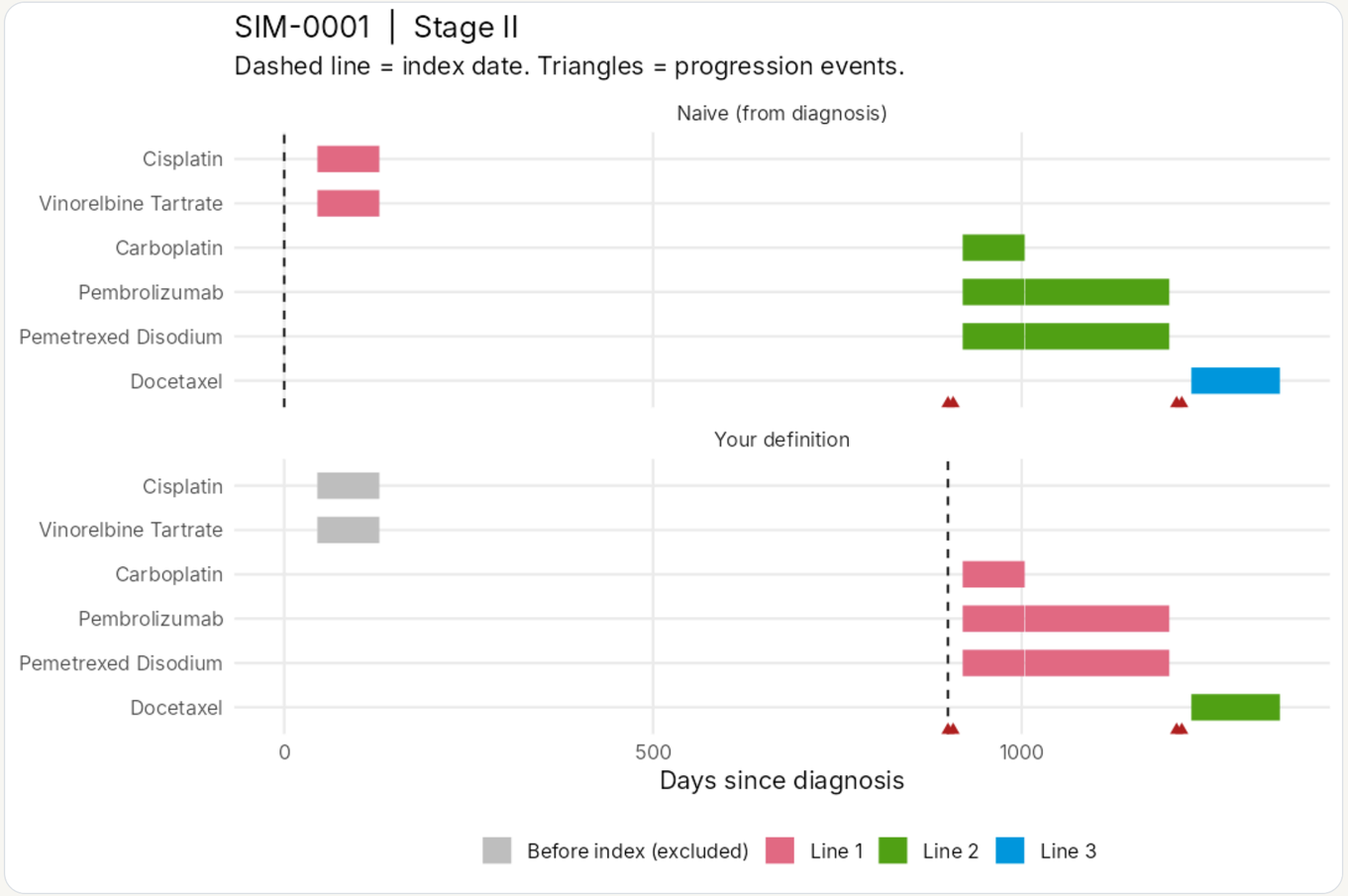
THE PROBLEM

Computers count the treatments wrong

Cancer treatment comes in rounds: **Line 1, Line 2, Line 3**.

When cancer is caught early, treated, then **comes back years later**, computers call the old treatment Line 1.

Every count after that is off by one.



Top: the old way. Bottom: genieLOT. One example patient (simulated, not real data).

WHAT WE FOUND · REAL LUNG CANCER DATA

2 in 3

patients whose cancer came back got the wrong Line 1 with the old way

63.7% · 198 of 311 patients · from 1,869 real cancer cases

Data: AACR Project GENIE, lung cancer (BPC NSCLC 2.0-public) · totals only, no individual patients

HOW WE KNOW IT WORKS

We checked our work three ways

100%

of the patients AACR itself marked as “cancer came back,” we found too

8 of 8

different ways of counting all show the same problem (53–67% wrong)

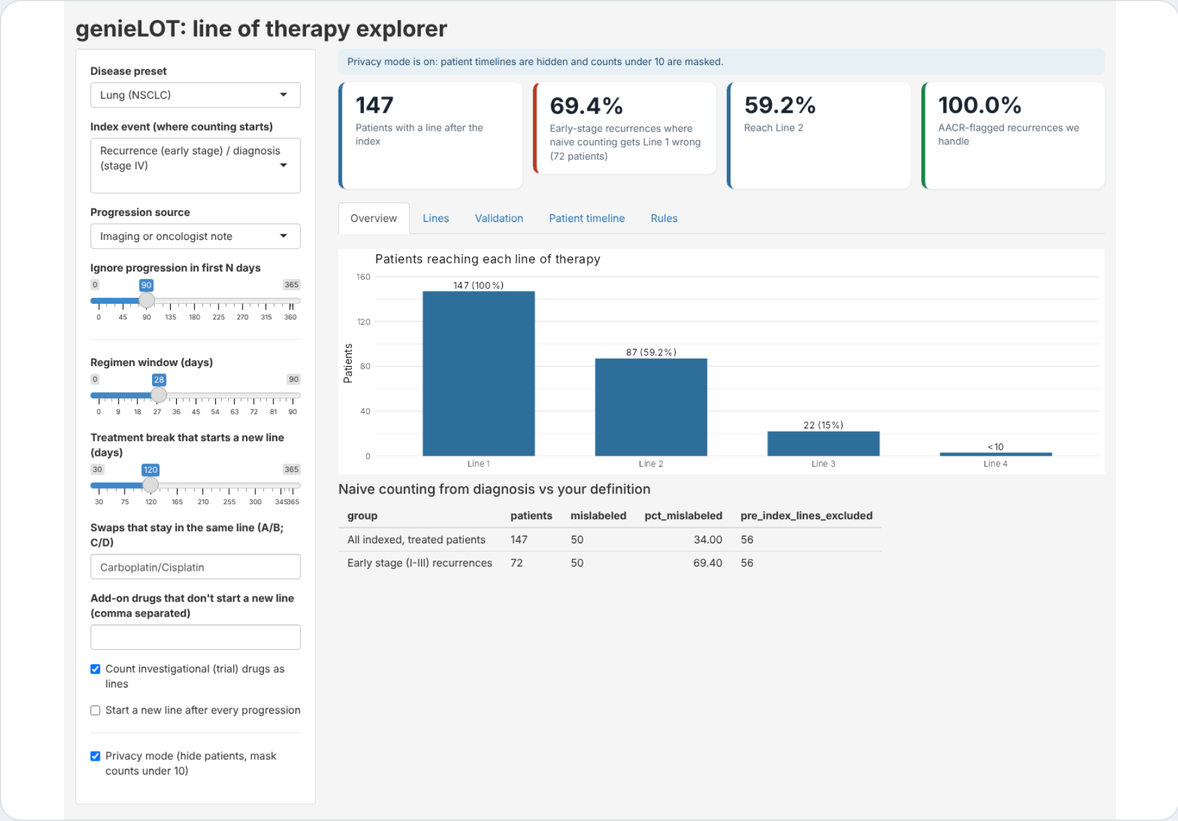
57

automatic tests check every rule, every time the code changes

AACR’s flag: 583 of 583 patients · details in backup slides and RESULTS.md on GitHub

ANYONE CAN USE IT

Free, open, and ready today



Pick where counting starts and set your own rules

Results update live in the app

Privacy mode keeps patients hidden

`github.com/Pchar046/genieLOT`

Results are based in whole or in part on data generated by the AACR Project GENIE Consortium (BPC NSCLC 2.0-public), accessed through the University of Miami. Screenshot uses simulated patients.